Sensory Perception



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Sensory Alterations

- Humans have five basic senses
  - Sight
  - Smell
  - Hearing
  - Taste
  - Touch
- The inability to see, smell, hear, taste, or feel are referred to as sensory alterations.



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Three Common Sensory Alterations

- Sensory Deficits
- Sensory Deprivation
- Sensory Overload



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A sensory deficit occurs when a client has a deficit in the expected function of one or more of their five senses. For example, a client may not be able to receive certain stimuli that allows them to process sounds below a certain decibel level, leading to a hearing deficit.

Sensory deficits can occur due to injury, illness, or aging and can have significant effects on a client's general well-being and daily functioning.

Sensory deprivation occurs when there is a reduction in or absence of stimuli to one or more of the five senses.

An example of sensory deprivation is blindness, which causes an absence of visual sensory stimulation.

Sensory overload is defined as receiving stimuli at a rate and intensity beyond the brain's ability to process the stimuli in a meaningful way. This can cause the client to feel anxious, restless, and confused.

For example, a client who is critically ill and frequently in pain may be receiving multiple sensory stimuli, both from their own body and the busy care environment: room lights, noises from monitors and other equipment, and frequent visits by nurses and providers.

Sensory Deprivation

Sensory Deprivation:

- Insufficient meaningful stimulation — common in hospitalized patients
  - Who is at risk: ICU patients, patients in private rooms with limited visitors, cognitively impaired older adults, patients with prolonged bedrest
  - What it looks like: Confusion, anxiety, hallucinations, depression, altered time perception
- Nursing interventions:
  - Encourage family presence and meaningful interaction
  - Provide clock and calendar — orient to day and time regularly
  - Encourage meaningful activity — TV, music, reading, conversation
  - Maintain day/night light cycles — lights up during the day, dimmed at night

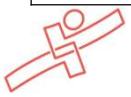


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ICU delirium — which has sensory deprivation and overload as contributing factors — affects up to 80% of mechanically ventilated patients and is associated with longer ICU stays, worse cognitive outcomes, and higher mortality. The ABCDEF bundle (Assess, prevent and manage pain; Both spontaneous awakening and breathing trials; Choice of analgesia and sedation; Delirium assessment; Early mobility; Family engagement) specifically addresses sensory factors. Even in a general hospital setting, the nurse who actively manages stimulation — maintaining day/night cycles, clustering care, reducing alarm noise — is preventing delirium and improving patient outcomes.

Deficit vs Deprivation

	Sensory Deficit	Sensory Deprivation
What it is	A problem with the sensory organ or pathway	A reduction or absence of meaningful stimulation
Where the problem lives	Inside the person — the receiver is impaired	In the environment — the signal is missing or reduced
Cause	Injury, illness, aging, or neurological damage	Isolation, restricted environment, or as a result of a sensory deficit
Example	Hearing loss, blindness, peripheral neuropathy	Quiet ICU room, no visitors, no clock, no windows
Nursing focus	Compensate for the loss — assistive devices, adapted communication, safety	Increase meaningful stimulation — orientation, interaction, day/night cycles



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Sensory deficit → often leads to → sensory deprivation
A patient with hearing loss (deficit) stops receiving meaningful auditory input from their environment (deprivation). A patient who is blind (deficit) is deprived of all visual stimuli (deprivation). The deficit is the broken receiver. The deprivation is what the brain experiences as a result.

Sensory Overload

Sensory Overload:

- Excessive or uncontrollable stimulation — also common in hospitalized patients
  - Who is at risk: ICU patients, post-operative patients, patients with pain, patients with anxiety or PTSD
  - What it looks like: Confusion, agitation, inability to rest, heightened pain perception, anxiety
- Nursing interventions:
  - Consolidate nursing care to reduce interruptions
  - Silence unnecessary alarms — respond promptly to reduce alarm fatigue for patient
  - Use a quiet, calm tone of voice
  - Offer eye mask and earplugs at night
  - Close the door when possible



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Hearing

- Hearing loss
  - Can occur at any point across lifespan
  - May be due to infection, injury, genetics, or age
  - Also can be from exposure to loud noises
  - Three types of hearing loss
    - Sensorineural – most common
    - Presbycusis – age related
    - Conductive – trauma, inflammation, ear wax impaction, perforated eardrum, blockage
  - Speak loud but in low tone
- Tinnitus
- Ototoxicity



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Vision Impairment

- Most common sensory deficit in older adults — and frequently underassessed

- Types:
  - Presbyopia
    - age-related loss of near vision (reading)
  - Cataracts
    - clouding of the lens from clumping of protein; gradual vision blurring
  - Glaucoma
    - increased intraocular pressure damaging the optic nerve; peripheral vision lost first
  - Macular degeneration
    - central vision loss; common in older adults
  - Diabetic retinopathy
    - damage from chronic hyperglycemia



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Vision impairment significantly increases fall risk in hospitalized patients. A patient who cannot see the IV pole, the edge of the bed, or the call light button is at immediate safety risk. The clock position system for describing food orientation is a practical, dignity-preserving technique that students can use immediately. Patients are often reluctant to report vision difficulty — the nurse who assesses proactively, particularly with older adults and diabetic patients, will identify needs that would otherwise go unmet.

Vision Impairment

- Nursing interventions:
  - Orient patient to room layout verbally and physically on admission
  - Describe food on tray using clock positions: "potatoes at 3 o'clock, green beans at 9 o'clock"
  - Ensure adequate lighting
  - Use large print for written materials; offer verbal alternatives
  - Keep call light, glasses, and personal items within reach
  - Do not rearrange the room without informing the patient
  - Ambulate safely
  - Assess glasses: clean lenses, proper fit, patient has them available



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Aphasia

- Expressive (nonfluent)
- Receptive
- Global



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Expressive
Inability to name common objects or express ideas in words or writing

Also known as nonfluent aphasia and is due to damage to the Broca area located in the frontal lobe of the brain. A client who has nonfluent aphasia may understand speech but be unable to speak the words they want to say. The client may speak in short phrases that the listener can understand, such as “hungry now”. Clients who have expressive aphasia are often easily frustrated as they are aware of their speech difficulty.

A client who has comprehensive aphasia speaks in long sentences that have no meaning and often include unnecessary and made-up words. Comprehensive aphasia also causes a client to not understand what is being said and they are unable to comprehend that others can’t understand what they are saying.

Receptive
Inability to understand written or spoken language

Global
Inability to understand language or communicate orally

causes severe impairments in communication. Global aphasia is caused by significant damage to various language areas of the brain. Clients who have this type of aphasia have poor comprehension of language and may be unable to form words or sentences.

Patients with aphasia have varied degrees of inability to speak, interpret, or understand language.
Expressive aphasia, a motor type of aphasia, is the inability to name common objects or express simple ideas in words or writing. For example, a patient understands a question but is unable to express an answer.
Receptive aphasia is the inability to understand written or spoken language. A patient is able to express words but is unable to understand questions or comments of others.
Global aphasia is the inability to understand language or communicate orally.

Aphasia: Types and Communication Approaches

- **Expressive Aphasia (Broca's):**
  - Patient understands language but cannot produce words fluently
  - Speech is halting, effortful, non-fluent
  - Patient is often acutely aware of the deficit and frustrated
  - Example: *"Pain... leg... bad"* (trying to say "I have bad pain in my leg")
  - Approach: Yes/no questions, picture boards, letter boards, patience (never finish their sentences)

  - https://www.youtube.com/watch?v=JWC-cVQmEmY



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The most critical point about aphasia is that it is a language disorder, not a cognitive disorder. A patient with expressive aphasia may be fully oriented and cognitively intact but unable to express it — treating them as cognitively impaired is both clinically inaccurate and harmful to dignity. Speech therapy should be involved early. The nurse's role includes documenting which communication strategies work for that specific patient and communicating this to every team member so the patient doesn't have to re-establish communication methods with each new caregiver.

Aphasia: Types and Communication Approaches

- **Receptive Aphasia (Wernicke's):**
  - Patient produces fluent speech that may not make sense
  - Patient has difficulty understanding spoken or written language
  - *"I was going to the blanket table before the curtain came in"* (fluent but nonsensical — called word salad)
  - *"You know that smoodle pinkered and then I want to get the help for the freezles, but it doesn't make any track."* (fluent but contains made up words)
  - Following commands may be inconsistent or absent
  - Approach: Simple short sentences, gestures, demonstration — verbal reassurance may not be comprehended

  - https://www.youtube.com/watch?v=3oef68YabD0



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Aphasia: Types and Communication Approaches

- **Global Aphasia:**
  - Both expression and comprehension severely impaired
  - Most severe form — typically from large dominant hemisphere stroke
  - Example: may repeat one word (yes, da, etc.) or may be completely silent
  - Approach: Nonverbal communication, consistent caregivers, calm environment, picture-based tools

- **For all aphasia types:**
  - Speak at a normal pace — do not shout
  - Involve speech therapy early
  - Ensure cognitive assessment accounts for aphasia — patient may be cognitively intact
  - Document the type of aphasia and what communication strategies are effective
  - Avoid saying “I know what you mean” unless you truly do!



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Touch

- Tactile hypersensitivity
- Tactile hyposensitivity
- Peripheral neuropathy
- Diabetic neuropathy



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A client who has tactile hypersensitivity has a peripheral and/or central nervous system that is overly sensitive to sensory input, specifically tactile stimuli. The client may be unable to wear clothing that has a zipper or a shirt that has long sleeves.

A client who has tactile hyposensitivity may not be able to feel extremes in temperature or have a limited ability to feel pain from bone fractures or lacerations. These clients may seek out tactile stimulation by frequently touching objects in their environment.

Peripheral neuropathy occurs following damage to or malfunction of a peripheral nerve. Peripheral neuropathy most often affects the feet and hands, but it can occur in other parts of the body. Manifestations include weakness, numbness, burning, tingling, pain, inability to detect temperature, and hyporeflexia. Two other types of neuropathy include idiopathic neuropathy and diabetic neuropathy.

Diabetic neuropathy is nerve damage that occurs in clients who have diabetes mellitus. Diabetic neuropathy occurs over time from high blood

glucose levels and high levels of triglycerides in the blood, which can cause damage to the nerves and to the small blood vessels supplying blood to the nerves.

Interventions

- Orient client to environment
- Communication aids
- Control stimuli
- Safety



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Communication Adaptations

- **Hearing Impairment:**
  - Face the patient directly — lip reading is a key compensatory strategy
  - Speak clearly at a normal pace — shouting distorts lip patterns and does not help sensorineural loss
  - Reduce background noise before speaking
  - Ensure hearing aids are in and functioning
  - Provide written instructions for all key information
  - Use interpreter services for sign language when needed — never use family as interpreter for clinical information

- **Vision Impairment:**
  - Always identify yourself when entering the room: "Mrs. Park, it's your nurse Jamie"
  - Describe what you are doing before you do it
  - Use large print materials; offer to read written materials aloud
  - Ensure glasses are clean and available before patient needs them



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The hearing aid maintenance point is practically important and often overlooked: a hearing aid that is not functioning is no better than no hearing aid. Wax guards clog, batteries die, and devices need cleaning. The nurse who checks hearing aid function as part of the admission assessment and ensures the patient has batteries available will significantly improve the patient's ability to participate in their own care. The instruction never to use family members as clinical interpreters is a patient safety and legal standard — family members may omit, soften, or alter information, and the dynamic of a family member interpreting medical information creates ethical complications around privacy and consent.

Questions?



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